

THE MAKING AND KNOWING PROJECT

Craft and Science in a Sixteenth-Century Artist's Manuscript: An Open-Access Digital Edition and Translation of BnF Ms. Fr. 640

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Introduction

In the last decades of the sixteenth century, an anonymous French-speaking craftsman took the unusual step of setting down on paper his techniques for a number of processes that we would now classify as part of the fine arts, of craft, and of technology. The processes included drawing-instruction, pigment-making, metal-coloring, counterfeit gem production, life-casting in metal, cannon-casting, tree-grafting, land-surveying, a practice of taxidermy to manufacture monstrous composite animals (kittens and bats), making paper *mâché* masks, and much more. The manuscript testifies to the widespread interest in processes of making art, as well as to the constant experimentation with natural materials undertaken in early modern workshops. Incorporated first into the library of the Dukes de Béthune and then into the Royal library (in 1622), this large, 170-folio manuscript now resides in the Bibliothèque nationale in Paris as Ms. Fr. 640, and has been digitized by <http://gallica.bnf.fr/ark:/12148/btv1b10500001g.r=.langEN>.

The project described below aims to produce an open-access digital edition and English translation of this intriguing text. The digital edition is only one dimension of this project, however, for the *process* by which this critical edition will be produced is as important as its product. Research for the edition will form an experiment in both pedagogy and humanistic research. It will involve students working alongside academic and museum-based historians of art and historians of science, in collaboration with experienced makers, to reconstruct the technical recipes contained in the manuscript. Their findings will be used to understand and annotate the digital edition, and their experiences will foster the sharing of expertise across disciplines as well as the engagement on the part of students with the material culture of the past.

The Manuscript

BnF Ms. Fr. 640 resembles most closely “books of secrets” that began circulating in the Middle Ages and then were printed in large numbers from the last decades of the fifteenth century. Although they sound esoteric, they are, in effect, books of techniques, which, while practical and useful, also hint at the tacit dimension and initiation process of

much craft knowledge. BnF Ms. Fr. 640 differs significantly, however, from other such collections of techniques in its apparent lack of formulaic recipes, its constant reference to the writer's own experiences, its extensive observations of animal behavior, and its sketched illustrations of technical points. The margins of the instructions are filled with the practitioner's comments on his experiments with different materials and procedures, an indication that the manuscript was most likely a set of working notes. The anonymous author-craftsperson is a vocal and self-reflective witness to his culture, and his text offers insight into how natural materials and art objects were collected, appreciated, and circulated in a period of burgeoning production and consumption of material goods. BnF Ms. Fr. 640 provides a rare view into the experience and significance of experimenting with and improving artistic techniques, as well as into attitudes to natural materials and processes.

Art historians, as well as historians of science and of material culture have recently become interested in the genre of early modern recipes, books of secrets, and how-to manuals because they shed light on a remarkable transition in European society, when previously illiterate craftspeople began writing down their working procedures. These technical books, printed in large numbers and through many editions between the 1490s and 1800, are now seen to have great significance in early modern Europe. Recent scholarship on recipes and technical writing has drawn attention, for example, to the importance of "makers' knowledge," and has illuminated the broader epistemological significance of art in early modern culture. Moreover, these technical manuals are now viewed as having fostered new attitudes to the creative process and to technical ability. BnF Ms. Fr. 640 seems to be unique among such technical writings. Other such manuals lack the immediacy, self-reflexivity, and process-oriented character of Ms. Fr. 640. This unusual document thus gives a remarkable view into the sixteenth-century artist's workshop, allowing scholars glimpses of the methodical experimentation undertaken in the production of art, the obviously laborious process of translating workshop procedures into words on paper, and the author-practitioner's taken-for-granted stance that his art-making functioned as a tool for the investigation of nature.

The procedures in this craft text seem not to be meant to be reproduced solely through the act of reading, but rather to be an invitation to imitate and experiment. By taking up this invitation and reconstructing the techniques contained within it, the proposed critical edition will make accessible an exceptionally intriguing and important source that will significantly enhance the corpus of early modern recipe collections, and also allow readers to understand and analyze the act of making art as the creation of knowledge.

The Project

The process of translating a technical treatise offers special challenges. The meaning of terms for materials, and sequence and process of techniques often cannot be determined simply by translating the words of the manuscript. This is especially true of Ms. Fr. 640 in which it is clear that the original author faced many challenges, first as he translated his own making and doing into words—by no means an easy process, as the looping and circuitous descriptions in Ms. Fr. 640 give ample evidence—then made a fair copy of those recipes and instructions, and finally, replicated many processes (indicated by the

extensive marginal notations describing further experimentation in the manuscript). Understanding of such a detailed and complex technical manual requires trying and testing—reconstructing—the instructions contained in the manuscript.

The proposed digital edition of Ms. Fr. 640 will form the center of an experiment in collaboration and pedagogy that explores the value of hands-on experience as a form of research and learning, not just for makers, art conservators, and artists, but also, uniquely, for students of the humanities. Historians, paleographers, makers, art and technical experts, and conservation and materials scientists will teach and learn together as they collaborate on researching the manuscript. All activities will take place primarily at Columbia University, however teaching and research partnerships will be implemented with Colour Context projects at the Max Planck Institute for the History of Science in Berlin, and with the Conservation Training Programs at the University of Amsterdam and the Cologne Fachhochschule, and with Dr. Erma Hermens, Lord Kelvin Adam Smith Associate Professor, and Dr. Mark Richer, Lecturer, of the Technical Art History Group, School for Culture and Creative Arts, College of Arts, University of Glasgow, as well as with the V&A/RCA History of Design Postgraduate Programme, and the Bard Graduate Center Graduate program in History of Design and Material Culture.

The Project has three main components:

1) Summer Paleography Workshops

Professor Marc Smith, Professeur de paléographie médiévale et moderne, Ecole des chartes, Paris, will lead workshops at Columbia University for advanced graduate students that will assist in revising and refining the existing transcription and translation of the complete manuscript. Pamela Smith commissioned a very rough transcription and translation of the entire manuscript, completed over the past nine years. These workshops will provide graduate students in French literary studies, history, and art history with training in Renaissance French hands, and the students will gain practice in correcting the transcription and translation of the manuscript. The first of these workshops, with 15 PhD students from North America and Europe, took place with tremendous success on June 9-27, 2014. Two more are planned for the following years.

2) Laboratory Seminars

Hands-on research into, and reconstruction of, the materials and techniques described in the manuscript will be carried out through seminars held in a laboratory at Columbia University, led by Seth Low Professor of History, Pamela Smith. The prototype for this course is the extensive reconstruction of the approximately 45 folios devoted to life casting in Ms. Fr. 640 that Professor Smith and metals conservator and silversmith Tonny Beentjes undertook in the conservation laboratory facilities of the Rijksmuseum, Amsterdam, between 2007 and 2011. The Columbia University Laboratory Seminars will be open to graduate students from Columbia and its consortial partners in the first year (2014-15) and to undergraduates thereafter. The course will include discussions of primary and secondary sources drawn from material culture studies, anthropology, history of science and technology, and art history, as well as hands-on work in the laboratory. Using the transcription and translation produced in the Paleography

Workshops, the Laboratory Seminars will focus each year on a single set of related techniques described in the manuscript. In 2014-15, the focus will be portrait medal casting, sand casting, and box moulding; in 2015-16, pigment making and painting; and in 2016-17, varnish making. Work in the laboratory each semester will begin with two-week residencies by expert makers. These experts will be artists, conservators, or craftspeople who have studied historic techniques in areas relevant to the manuscript. Their expertise will provide a crucial link between the text, students, and scholars working in the laboratory, and their contributions will be of invaluable practical value in performing reconstructions, and in the critical annotation of the manuscript.

The process of translation from words into actions, and the multiple issues it raises will provide valuable lessons about translation itself (in all its many meanings in this project), about experiential knowledge, and about methodologies of research in history to the students in the courses, and, more generally to scholars in multiple fields. The participants' grappling with these issues will be reflected in electronic laboratory notebook entries, images, and video footage as part of the unorthodox visual annotations in the digital edition of the manuscript. An electronic platform being developed in consultation with the Digital Humanities Center at Columbia University Libraries will facilitate the group work of transcription and translation, as well as allow students and Visiting Expert Makers to record the results in words and images of their work in the laboratory.

3) Working Groups

The three working groups include Metals and Metalworking; Painting and Pigments; and Natural History and Mechanics (final participant group is still being composed) and will be comprised of scholars and technical experts organized according to these different topics contained in the manuscript. In May of each project year, the students in the Laboratory Seminars and the members of the relevant Working Groups will hold a three-day conference on issues related to the year's focus. Members of the Working Groups will have access to the electronic platform so they can follow the progress of transcription, translation, and annotation of the manuscript, and they will be invited to contribute to the annotation of the manuscript.

Dissemination and Impact

The open-access digital publication of a scholarly edition, transcription, and translation of Ms. Fr. 640 will serve many communities of scholars:

- It will provide many different types of historians in science museums and art museums as well as in the academy with a unique primary source that will become a point of reference for any historian seeking information on technical procedures and attitudes to nature in early modern Europe.
- The materials-based research will be especially useful to technical art historians, conservators, and museum scholars specializing in Renaissance Europe, who can use either the findings or the methods communicated in the publication to inform their own research.

- The light the manuscript sheds on how and why nature was investigated, how natural materials and natural objects were used in art, and collected and appreciated in early modern Europe will be of deep interest to historians of art, science, technology, and material culture.
- The edition will also be designed to be navigable by the broader public, who will be able to access the publication freely.

The project's methodology can also serve as a blueprint for hands-on historical research and new methods for codifying these kinds of findings. BnF Ms. Fr. 640 is only one of numerous technical recipe collections that could be investigated in this project, and the project foresees moving on to the study of other collections after the first years spent studying, reconstructing, and preparing BnF Ms. Fr. 640 for publication. Through the involvement of students, and by means of the conferences, workshops, and field experiences, this project will also foster further research on technical recipes and procedures, on the connections between processes of making and their codification in written texts, and, more generally, between making in the workshop or laboratory and the knowledge, or science, it produces.

Progress to date:

1. A rough transcription and translation of BnF Ms. Fr. 640 has been completed, digitized and transferred to a collaborative digital workspace. A little more than one-third of this version was revised, reworked, and, in many cases, transcribed and translated from scratch by the June 2014 Paleography Workshop. The 2014 Workshop concentrated on the pages of the manuscript that contain metalworking recipes, in order to prepare these recipes for the reconstruction work in the laboratory beginning in September 2014.
2. An Advisory Committee for the project has been formed (available on request).
3. Columbia University has committed to providing space for the Paleography Workshops and the Annual Working Group Meetings, as well as providing a laboratory for the Laboratory Seminars. Three postdoc positions have been created to teach half-time in the Laboratory Seminars and half-time in the Core Curriculum. These individuals have both a PhD in history or history of science and lab or studio-based experience.
4. A three-year NSF Grant SBES Directorate #1430843 was obtained for partial support of the project from 9/1/14-8/30/17. Applications have been submitted to private foundations and the NEH for support of various components of the Project.
5. In 2007-11, supported in part by a Kress Paired Fellowship at CASVA and by a Mellon New Directions Fellowship, Pamela Smith and Tonny Beentjes (Programme Leader of Metalwork, University of Amsterdam) carried out reconstructions of the procedures for life-casting described in BnF Ms. Fr. 640, resulting in the following publications:
 - Tonny Beentjes and Pamela H. Smith, "Sixteenth-century life-casting techniques: experimental reconstructions based on a preserved manuscript," *The Renaissance*

- Workshop*, ed. by David Saunders, Marika Spring and Andrew Meek (London: Archetype Publications Ltd, 2013), 144-51.
- Pamela H. Smith, "Making Things: Techniques and books in early modern Europe," *Things*, Paula Findlen, ed. (London: Routledge, 2013), 173-203.
 - Pamela H. Smith, "In the Workshop of History: Making, Writing, and Meaning," *Shaping Objects: Art, Materials, Making, and Meanings in the Early Modern World*, an article series of *West 86th: A Journal of Decorative Arts, Design History, and Material Culture*, 19 (2012): 4-31.
 - Pamela H. Smith and Tonny Beentjes, "Nature and Art, Making and Knowing: Reconstructing Sixteenth-Century Life Casting Techniques," *Renaissance Quarterly*, 63 (2010): 128-179.